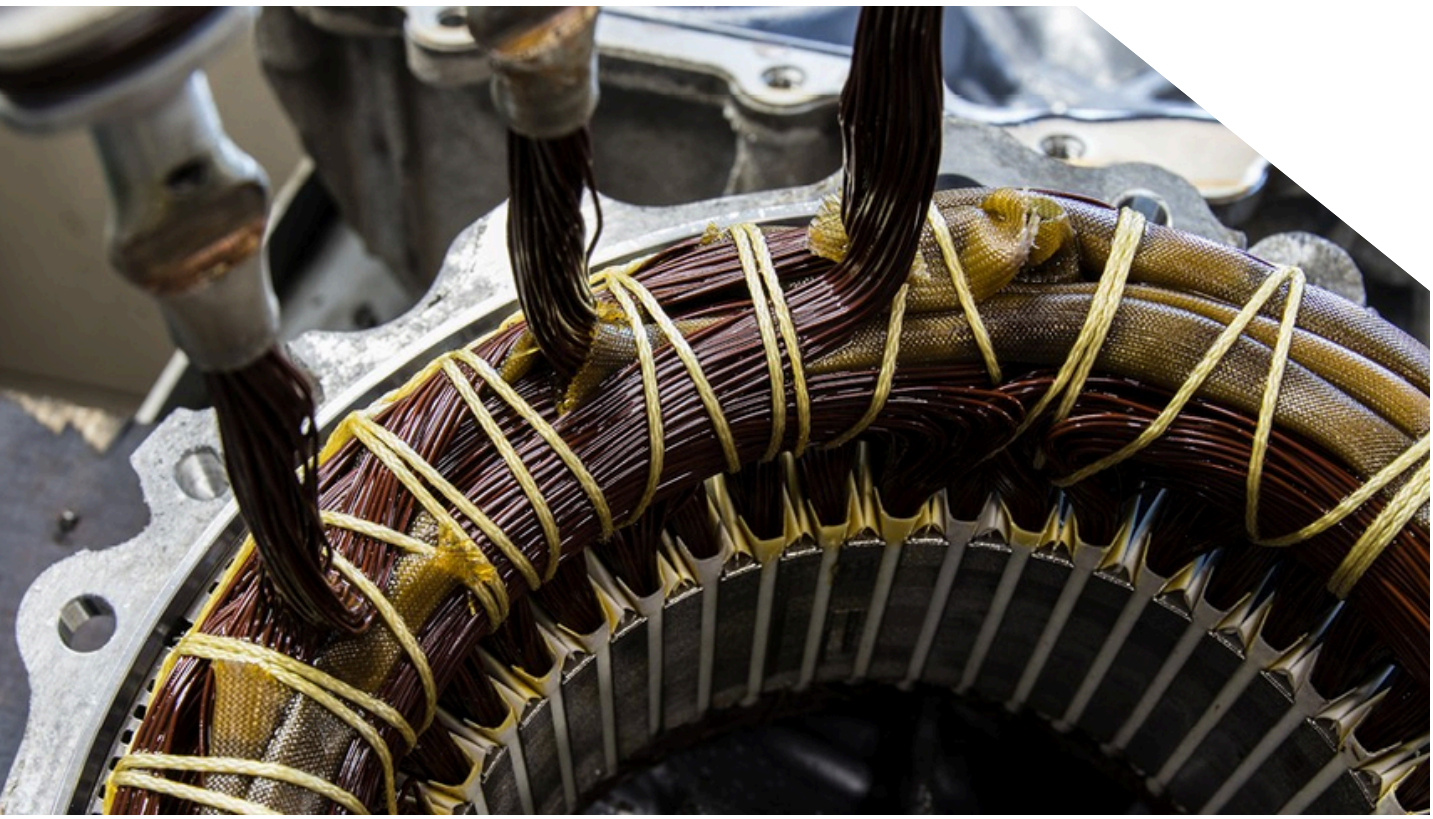
First Report

Mathematical Engineering
Curriculum in Financial Engineering
Stochastic Methods for Finance



ONE-STEP BINOMIAL OPTION PRICING REPORT

Valuation of 3-Month and 6-Month Call Options on MP Strategic Assets

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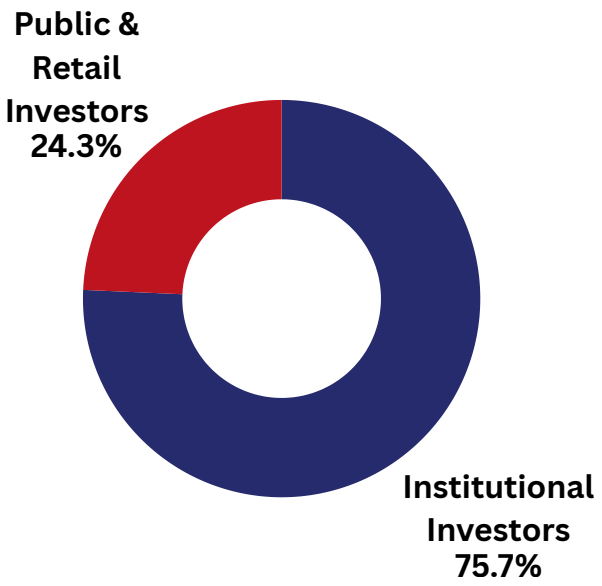
Company Profile:
MP Materials Corp. (NYSE: MP)

MP Materials is the largest producer of rare earth materials in the Western Hemisphere. The company owns and operates the Mountain Pass Rare Earth Mine and Processing Facility in California, a site of immense geopolitical and strategic importance. Their primary products, Neodymium and Praseodymium (NdPr), are essential components for the high-strength permanent magnets used in advanced defense systems, (EV) motors and industrial production.
As the world shifts toward a decarbonized economy, MP Materials stands as a critical node in the global supply chain, aiming to restore a full domestic "mine-to-magnet" capability in the United States. Given its exposure to commodity price fluctuations and the rapid growth of the EV market, the stock exhibits a high-volatility profile, making it a compelling subject for stochastic derivative modeling.

Ownership Structure (Holders)

The stock has significant institutional ownership from major asset managers like Vanguard, BlackRock, and State Street, ensuring strong market liquidity and stable trading volumes. This concentration supports reliable Bid/Ask spreads for evaluating the option's fair value using our binomial model.



Key Statistics & Market Data

The following metrics, captured as of March 2026, highlight the stock's suitability for stochastic modeling:

- Market Capitalization: **$10.16B** – Positions MP as a high-growth "Large-Cap" player in the strategic materials sector.
- 52-Week Price Range: **$18.64 – $100.25** – This extreme price excursion confirms the high historical volatility (utilized in our 3-month (69%) and 6-month (81%) models.
- Beta: 1.62 – A Beta significantly higher than the market average (1.0) reinforces the stock's aggressive risk/return profile and justifies the wide u and d factors.
- Average Daily Volume (3M): **7.47M shares** – Sustained high trading volume ensures that the underlying asset is liquid enough for efficient option pricing.
- 1-Year Change: **+113.2%** – The strong upward momentum, despite negative net income due to heavy CAPEX reinvestment, underscores a market driven by future growth expectations and volatility rather than current earnings.

0.1 Methodology: Step-by-Step Excel Implementation

The MP Materials (MP) Call option was valued in Excel using a One-Step Binomial Model calibrated on market and historical data.

Every step implementation can be checked in the provided file: `Report1_Quartucci_Davide.xlsx`

Step 1: Finding Data & Daily Returns Using Yahoo Finance, the Spot Price (S_0), the ATM Strike Price (K), and the Bid and Ask quotes of the selected 3-month and 6-month Call options were collected. In Excel, the market benchmark was computed as the Mid-Price:

$$P_{mid} = \frac{P_{bid} + P_{ask}}{2}$$

Step 2: Historical Data Extraction and Excel Integration To respect the *Matching Principle*, the volatility input was estimated on historical windows consistent with option maturity. Two datasets of daily closing prices (P_t) were downloaded from Yahoo Finance:

- A **3-month** historical sample.
- A **6-month** historical sample.

The CSV files were imported into Excel and stored in two separate worksheets, `3month` and `6month`, to keep the calculations distinct.

Step 3: Computation of Daily Returns In each Excel sheet, daily returns were computed from closing prices using:

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

where P_t is the closing price on day t. In practice, a dedicated return column was created and filled down over the full sample.

Step 4: Estimation of Historical Volatility (σ_D and σ_Y) Historical volatility was estimated in Excel as the sample standard deviation of returns, using `=STDEV.S()` to obtain daily volatility (σ_D). This was then annualized as:

$$\sigma_Y = \sigma_D \cdot \sqrt{252}$$

The resulting values used in the pricing sheet were:

- **3-Month Sample:** $\sigma_D = 4,35\% \implies \sigma_Y = 69,07\%$
- **6-Month Sample:** $\sigma_D = 5,14\% \implies \sigma_Y = 81,63\%$

The longer lookback period captured higher historical variability, producing a larger volatility input for the 6-month model.

Step 5: Calibration of Binomial Price Multipliers (u and d) Using the CRR parametrization, the up and down multipliers were computed in Excel from annualized volatility and time to maturity (Δt):

$$u = e^{\sigma_Y \sqrt{\Delta t}}$$

$$d = e^{-\sigma_Y \sqrt{\Delta t}} = \frac{1}{u}$$

Excel formulas based on `EXP()` and `SQRT()` gave:

- **3-Month Horizon ($\sigma_Y = 69.07\%$):** $u = 1.4125$ and $d = 0.7080$
- **6-Month Horizon ($\sigma_Y = 81.63\%$):** $u = 1.7810$ and $d = 0.5615$

The higher 6-month volatility implies a wider range of possible future prices.

Step 6: Risk-Free Rate (SOFR) and Simple Compounding Factors The risk-free rate (r) was proxied by the 3-month and 6-month Term SOFR. In line with the project requirements, simple compounding was used instead of continuous compounding:

$$\text{Capitalisation Factor} = 1 + r\Delta t$$

$$\text{Discount Factor} = \frac{1}{1 + r\Delta t}$$

These expressions were implemented directly in Excel and linked to the pricing block.

Step 7: Risk-Neutral Probabilities and Theoretical Pricing (P_0) Having zero dividend yield, the risk-neutral probability was computed as:

$$q = \frac{(1 + r\Delta t) - d}{u - d}$$

The terminal Call payoffs were then evaluated in Excel with `MAX()`:

$$f_u = \max(S_0 u - K, 0)$$

$$f_d = \max(S_0 d - K, 0)$$

Finally, the theoretical price was obtained by discounting the expected payoff:

$$P_0 = \frac{1}{1 + r\Delta t}[q f_u + (1 - q) f_d]$$

The same Excel structure was replicated for both maturities, and the resulting P_0 values were compared with market Mid-Prices.

0.2 Visual Representation: The Binomial Tree Structure

Before examining the numerical results, it is instructive to visualize the geometry of the price process we have modeled. Figure 1 presents a stylized one-step binomial tree. The model assumes that from the current Spot Price (S_0), the asset price can only move to two possible states at maturity: an 'Up' state ($S_0 \cdot u$) and a 'Down' state ($S_0 \cdot d$).

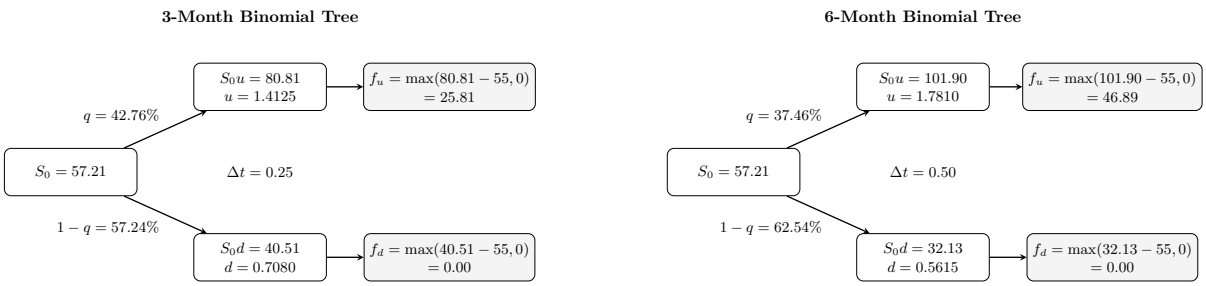


Figure 1: Personalized one-step binomial trees for the 3-month and 6-month MP Materials Call valuation in Excel.

0.3 Results Table & Critical Conclusions

The comparative analysis provides powerful insights into the mechanics of option pricing and the structural limitations of the One-Step Binomial Model when applied to high-volatility equities like MP Materials.

0.3.1 Confirmation of the Time Value Principle

The empirical results clearly demonstrate the fundamental principle of *Time Value*. By extending the option's maturity from 3 to 6 months, the theoretical premium (P_0) increases significantly from \$10.93 to \$17.25. A longer time horizon grants the underlying asset more time to experience favorable volatility, which is geometrically reflected in the massive increase of the Up-Node payoff (f_u), soaring from \$25.81 (3-month) to \$46.89 (6-month).

Parameter	3-Month Analysis	6-Month Analysis
Underlying & Market Data		
Spot Price (S_0)	\$57.21	\$57.21
Strike Price (K)	\$55.00	\$55.00
Market Bid / Ask	\$9.45 / \$9.60	\$12.25 / \$13.25
Market Mid-Price	**\$9.53**	**\$12.75**
Model Inputs		
Annualized Volatility (σ_Y)	69.07%	81.63%
Risk-Free Rate (SOFR)	3.68%	3.66%
Capitalisation Factor	1.0092	1.0183
Discount Factor	0.9909	0.9820
Binomial Dynamics		
Up Factor (u) / Down Factor (d)	1.4125 / 0.7080	1.7810 / 0.5615
Risk-Neutral Prob. (q / $1-q$)	42.76% / 57.24%	37.46% / 62.54%
Payoff Up (f_u) / Payoff Down (f_d)	\$25.81 / \$0.00	\$46.89 / \$0.00
Valuation & Spread		
Theoretical Price (P_0)	**\$10.93**	**\$17.25**
Pricing Gap (Model Spread \$)	+\$1.41	+\$4.50
Pricing Gap (Model Spread %)	**+14.79%**	**+35.30%**

Table 1: Consolidated Output: One-Step Binomial Model vs. Market Pricing.

0.3.2 Divergence: Historical vs. Implied Volatility

The most critical finding of this report is the systematic overvaluation of the options by the theoretical model, characterized by a positive pricing gap of +14.79% for the 3-month contract and a substantial +35.30% for the 6-month contract.

This discrepancy highlights the fundamental difference between backward-looking and forward-looking risk metrics. The model was calibrated using **Historical Volatility** (σ_Y of 69.07% and 81.63%), capturing a period of extreme market turbulence for MP Materials (evident from its wide 52-week range). However, the real market Mid-Prices (\$9.53 and \$12.75) are driven by **Implied Volatility**. The lower market premiums indicate that institutional investors and market makers expect the future price action of MP Materials to stabilize, thereby pricing in a lower volatility regime than what was observed in the past.

0.3.3 Structural Limitations of the One-Step Framework

Finally, the widening of the error margin in the 6-month analysis (+35.30%) exposes the inherent limitations of a single-period binomial tree. Oversimplifying 6 months of continuous trading into a single discrete jump creates extreme, unrealistic terminal states. Specifically, the model assumes the stock will either surge by 78.1% ($u = 1.7810$) or crash by 43.8% ($d = 0.5615$) in exactly one step. In reality, option pricing over longer horizons requires multi-step lattices (e.g., 50 or 100 steps) or continuous-time models (Black-Scholes) to properly smooth the distribution of terminal prices and align theoretical values closer to market realities.